JavaScript Global Execution Context - Complete Notes

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1. Global Execution Context

What is Execution Context?

- **Definition**: Execution Context is the environment in which JavaScript code is executed
- **Global Execution Context (GEC)**: The default execution context where JavaScript code runs
- When JavaScript engine starts, it creates the Global Execution Context

Key Points:

- 📌 Every JavaScript program has at least one execution context (Global)
- 📌 It's created when the JavaScript engine starts
- 📌 It's the base/default context for all JavaScript execution

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2. Window Variable

What is the Window Object?

- **Window**: A global object created by the JavaScript engine
- **Browser Environment**: In browsers, `window` represents the browser window
- **Global Scope**: All global variables and functions become properties of the window object

Important Facts:

```
// These are equivalent in global scope:
var a = 10;
window.a = 10;

// Both will give the same result
console.log(a);        // 10
console.log(window.a); // 10
```

Key Points:

- 🌐 `window` is available globally
- 🌐 All global variables become properties of `window`
- 🌐 Functions declared globally also become methods of `window`

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3. JavaScript Code Run in Two Phases

JavaScript engine executes code in **TWO DISTINCT PHASES**:

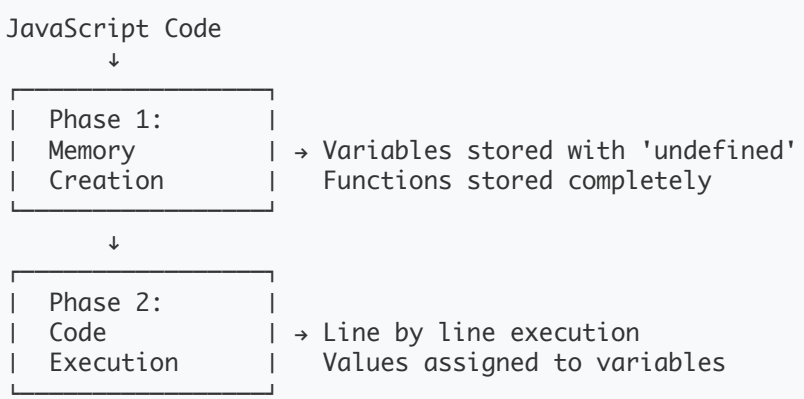
Phase 1: Memory Creation Phase (Variable Phase)

- Also called **"Hoisting Phase"**
- Memory allocation happens
- Variables and functions are registered

Phase 2: Execution Phase

- Actual code execution happens
- Values are assigned to variables
- Functions are called and executed

Visual Representation:



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4. Variable Phase (Memory Creation Phase)

What Happens in This Phase?

1. **Memory Allocation**: JavaScript engine scans through the entire code
2. **Variable Registration**: All variables are registered in memory
3. **Default Values**: Variables are assigned `undefined` initially
4. **Function Storage**: Complete function definitions are stored

Example:

```
// Before execution starts, in Memory Creation Phase:

var a = 10;         // a → undefined (initially)
let b = 20;         // b → undefined (initially)
const c = 30;       // c → undefined (initially)

function myFunc() {  // myFunc → complete function stored
    return "Hello";
}
```

Memory State After Variable Phase:

```
Global Memory:
├── a: undefined
├── b: undefined
├── c: undefined
└── myFunc: f() { return "Hello"; }
```

Important Notes:

- ⚠️ `var` declarations are hoisted and initialized with `undefined`
- ⚠️ `let` and `const` are hoisted but not initialized (Temporal Dead Zone)
- ⚠️ Function declarations are completely hoisted
- ⚠️ This is why we can call functions before they're declared!

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5. Execution Phase

What Happens in This Phase?

1. **Line-by-Line Execution**: Code runs from top to bottom
2. **Value Assignment**: Variables get their actual values
3. **Function Calls**: Functions are executed when called
4. **Context Updates**: Memory values are updated as code runs

Example Walkthrough:

```
// Original Code:
console.log(a);    // What will this print?
var a = 10;
console.log(a);    // What will this print?

function test() {
    return "Working!";
}
console.log(test()); // What will this print?
```

Step-by-Step Execution:

After Memory Creation Phase:

```
Memory:
├── a: undefined
└── test: f() { return "Working!"; }
```

During Execution Phase:

```
console.log(a);    // Prints: undefined (from memory)
var a = 10;        // Now a gets value 10
console.log(a);    // Prints: 10
console.log(test()); // Prints: "Working!"
```

Final Memory State:

```
Memory:
├── a: 10
└── test: f() { return "Working!"; }
```

🎯 Key Takeaways

Understanding Hoisting:

- **Variables**: Declared but not initialized (`undefined`)
- **Functions**: Completely available before declaration
- **Execution**: Happens in two distinct phases

Common Interview Questions:

Q: What will this code output?

```
console.log(x);
var x = 5;
console.log(x);
```

Answer:

- First `console.log(x)` → `undefined`
- Second `console.log(x)` → `5`

Q: Why can we call functions before declaring them?

Answer: Because of the Memory Creation Phase, function declarations are completely stored in memory before execution begins.

Best Practices:

- ✅ Always declare variables before using them
- ✅ Use `let` and `const` instead of `var` when possible
- ✅ Understand hoisting to avoid unexpected behavior

- ✅ Remember: Declaration ≠ Initialization

📝 Practice Examples

Example 1: Variable Hoisting

```
console.log(name); // ?
var name = "JavaScript";
console.log(name); // ?

// Output:
// undefined
// JavaScript
```

Example 2: Function Hoisting

```
sayHello(); // This works!
function sayHello() {
    console.log("Hello World!");
}
// Output: Hello World!
```

Example 3: Mixed Example

```
console.log(a); // ?
console.log(b); // ?
console.log(c()); // ?

var a = 1;
let b = 2;
function c() { return 3; }

// Output:
// undefined
// ReferenceError (Temporal Dead Zone)
// 3
```
